Efficient treatment and simultaneous energy recovery from cattle wastewater by micro-aerated UASB and SBR coupled system

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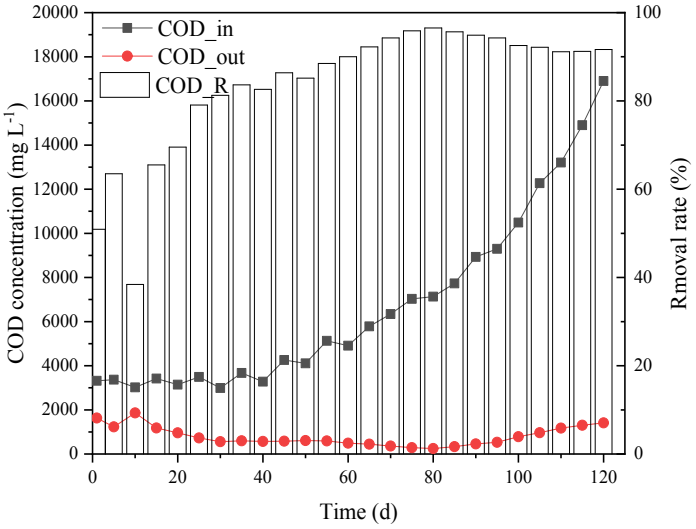
Supporting information content

Figure S1. Typical pollutant removal**:** (a) Concentration and removal efficiencies of COD; (b) NH_4^+-N; and (c) PO_4^{3-} in UASB without micro-aeration.

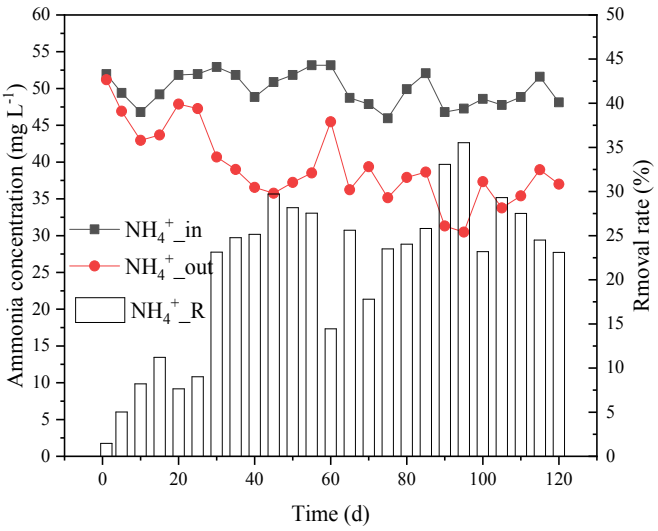
Figure S2. Nitrogen composition of UASB 1 and UASB 2 in a typical cycle.

Figure S3. The concentration variation and removal rate of COD of UASB under different optimization conditions: (a) Organic load; (b) HRT, and (c) C/N ratio.

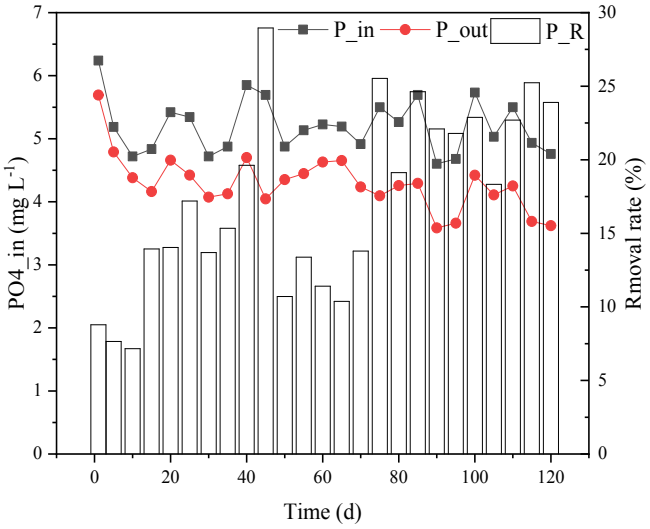
Figure S4. The concentration variation and removal rate of COD of SBR under different optimization conditions: (a) DO level; and (b) reaction time.


COD_in
COD_out
COD_R
COD concentration (mg L-1)
Rmoval rate (%)
Time (d)
0
2000
4000
6000
8000
10000
12000
14000
16000
18000
20000
0
20
40
60
80
100
120

(a)


NH4+_in
NH4+_out
NH4+_R
Ammonia concentration (mg L-1)
Rmoval rate (%)
Time (d)
0
5
10
15
20
25
30
35
40
45
50
55
60
0
20
40
60
80
100
120

(b)



(c)

Figure S1. Typical pollutant removal**:** (a) Concentration and removal efficiencies of COD; (b) NH_4^+-N; and (c) PO_4^{3-} in UASB without micro-aeration.

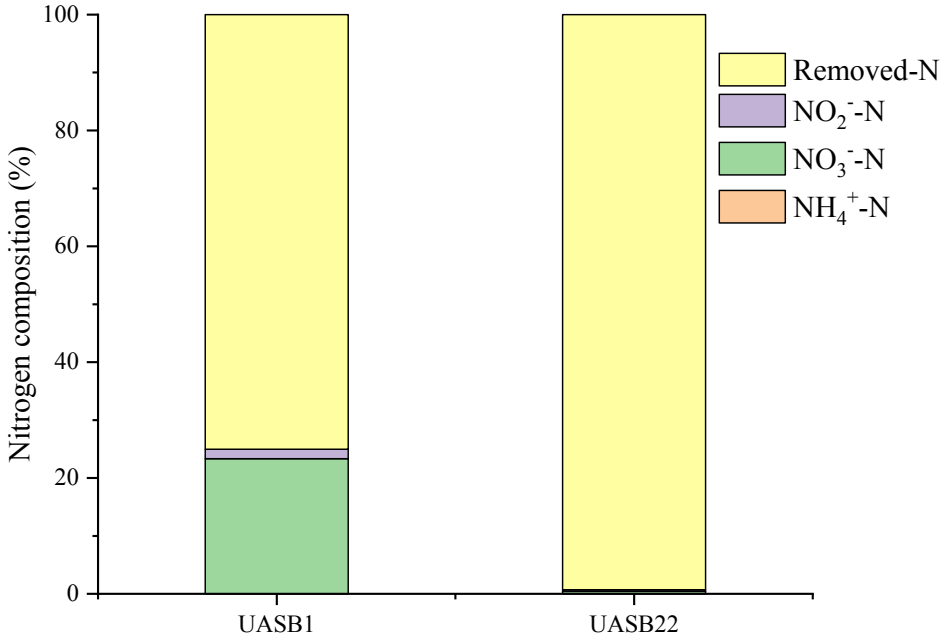
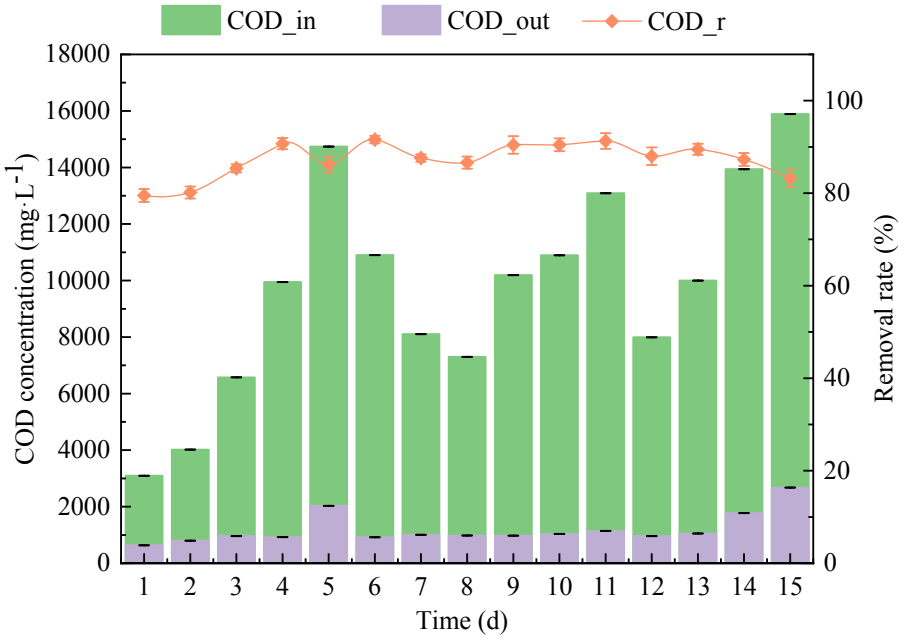
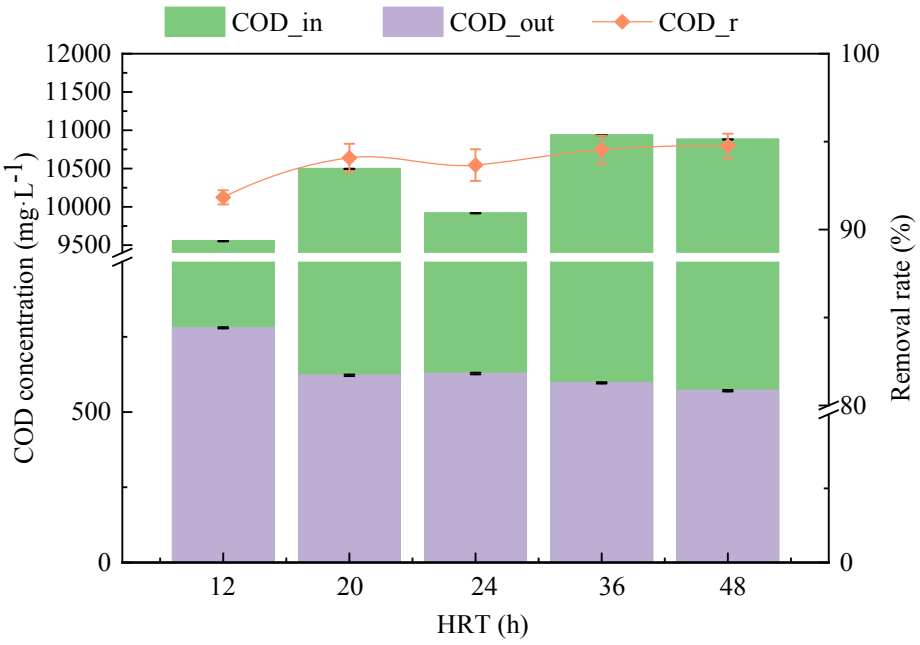


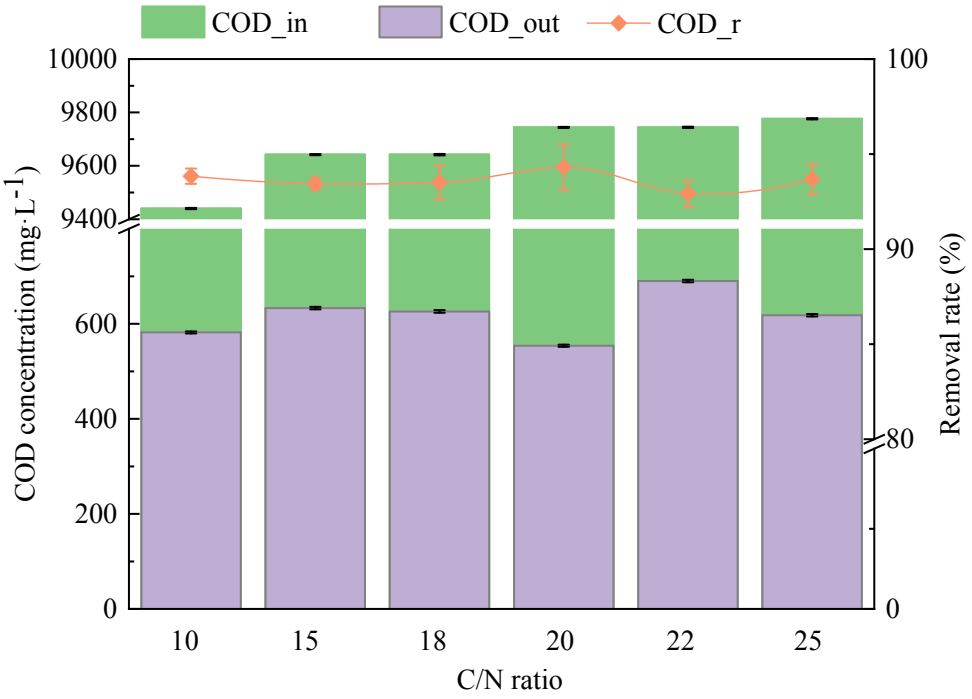
Figure S2. Nitrogen composition of UASB 1 and UASB 2 in a typical cycle.


COD_in
COD_out
COD_r
COD concentration (mg·L-1)
Removal rate (%)
Time (d)
0
2000
4000
6000
8000
10000
12000
14000
16000
18000
0
20
40
60
80
100
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15

(a)

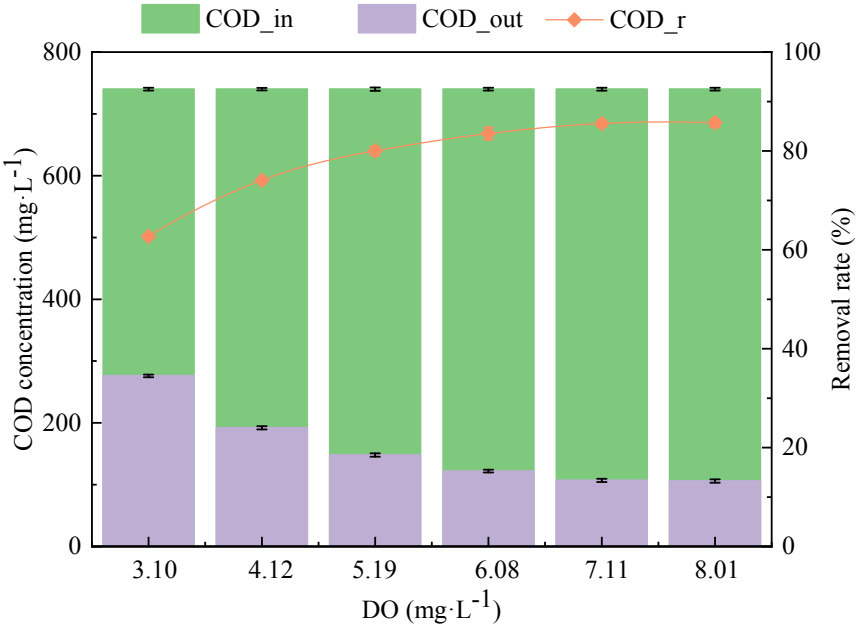

COD_in
COD_out
COD_r
COD concentration (mg·L-1)
Removal rate (%)
HRT (h)
9500
10000
10500
11000
11500
12000
500
0
80
90
100
12
20
24
36
48

(b)

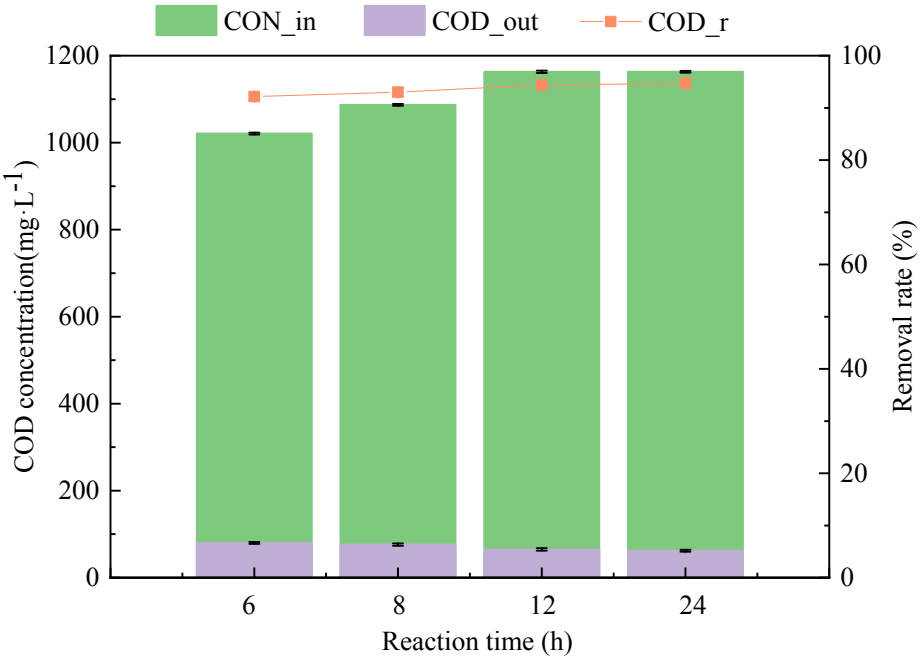


(c)

Figure S3. The concentration variation and removal rate of COD of UASB under different optimization conditions: (a) Organic load; (b) HRT, and (c) C/N ratio.



(a)



(b)

Figure S4. The concentration variation and removal rate of COD of SBR under different optimization conditions: (a) DO level; and (b) reaction time.